

Maximal Signed Volume for k -increasing Quasi-Copulas

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SPMS 26

Lecce, Italy, September 2026

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Outline

1. Quasi-copulas and k -increasingness

k -dimensional sections and volumes of boxes

2. Maximal signed volumes

Definitions of $MNV(d, k)$ and $MPV(d, k)$

3. Main results

Exact formulas, extremal boxes, and realizing quasi-copulas

4. Examples and asymptotic behavior

Dependence on d and k , including the case $k = 1$

5. Proof strategy

Symmetrization, finite differences, and primal-dual linear programming

Sections of a quasi-copula

A mapping $Q : [0, 1]^d \rightarrow [0, 1]$ is a d -variate **quasi-copula** if it is coordinatewise nondecreasing and

$$i) \ Q(x) = 0 \quad \text{if some } x_i = 0, \quad ii) \ Q(1, \dots, 1, x_i, 1, \dots, 1) = x_i, \quad iii) \ |Q(x) - Q(y)| \leq \sum_{i=1}^d |x_i - y_i|.$$

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k -dimensional section of Q :

1. Choose a k -element set $A \subseteq \{1, \dots, d\}$.
2. The coordinates indexed by A are allowed to vary.
3. The coordinates of A^c are fixed at some $a \in [0, 1]^d$.
4. The resulting restriction of Q is denoted by $Q_{a,A}$.

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Q is **k -increasing** if every section $Q_{a,A}$ has nonnegative volume on every k -box R :

$$V_{Q_{a,A}}(R) = \sum_{u \in \text{Vert}(R)} (-1)^{S_R(u)} Q_{a,A}(u) \geq 0.$$

Here $S_R(u)$ is the number of coordinates of u that are lower endpoints of R .

A 2-dimensional section in dimension 3

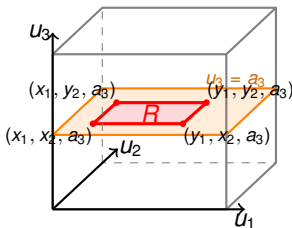
Fix $u_3 = a_3$. Then

$$Q_{a, \{1,2\}}(u_1, u_2) = Q(u_1, u_2, a_3),$$

$$R = [x_1, y_1] \times [x_2, y_2] \times \{a_3\}.$$

Here

$$x_1 \leq y_1, \quad x_2 \leq y_2.$$



The signed volume of R is

$$\begin{aligned} V_Q(R) &= Q(y_1, y_2, a_3) \\ &\quad - Q(x_1, y_2, a_3) \\ &\quad - Q(y_1, x_2, a_3) \\ &\quad + Q(x_1, x_2, a_3). \end{aligned}$$

The section is 2-increasing if

$$V_Q(R) \geq 0$$

for every rectangle R .

Main result: maximal negative volume

Let $d, k \in \mathbb{N}$ with $2 \leq k \leq d$.

$\text{MNV}(d, k)$ is the minimum of $V_Q(\mathcal{B})$, where:

- ▶ $\mathcal{B}$ ranges over all d -boxes in $[0, 1]^d$ and
- ▶ Q ranges over all k -increasing d -variate quasi-copulas.

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- ▶ Q ranges over all k -increasing d -variate quasi-copulas.

Theorem (Omladič, Vuk, Z., 2026b)

We have

$$\text{MNV}(d, k) = \begin{cases} 0, & k = d, \\ -\frac{1}{\binom{d-1}{k-1}} \max_{\substack{0 \leq r \leq d-k \\ r \equiv d-k-1 \pmod{2}}} \binom{d-1}{r}, & k < d. \end{cases}$$

Asymptotically, for a fixed k , the growth is exponential in d :

$$\text{MNV}(d, k) \sim -\frac{(k-1)!}{\sqrt{2\pi}} \frac{2^d}{d^{k-\frac{1}{2}}}, \quad d \rightarrow \infty,$$

Realization of the extremal value

Theorem (Omladič, Vuk, Z., 2026b)

Assume $2 \leq k < d$, and let r_0 be as on the previous slide.

Extremal box

$$a = \frac{k-1}{d-1-r_0}, \quad \mathcal{B} = [a, 1]^d.$$

For $z \in \text{Vert}(\mathcal{B})$, set

$$S(z) = \text{card}\{j : z_j = a\}.$$

Thus, $S(z)$ counts the lower coordinates of z .

These vertex values extend to a k -increasing quasi-copula $Q : [0, 1]^d \rightarrow [0, 1]$ satisfying

$$V_Q(\mathcal{B}) = \text{MNV}(d, k).$$

Vertex values

Define

$$Q(z) = \frac{\binom{d-S(z)-r_0-1}{k-1}}{\binom{d-1-r_0}{k-1}}, \quad z \in \text{Vert}(\mathcal{B}).$$

Vertices with the same number of lower coordinates receive the same value.

We use $\binom{n}{m} = 0$ whenever $n < m$.

Main result: maximal positive volume

Let $d, k \in \mathbb{N}$ with $2 \leq k \leq d$.

$\text{MPV}(d, k)$ is the maximum of $V_Q(\mathcal{B})$, where:

- ▶ $\mathcal{B}$ ranges over all d -boxes in $[0, 1]^d$,
- ▶ Q ranges over all k -increasing d -variate quasi-copulas.

Theorem (Omladič, Vuk, Z., 2026b)

We have

$$\text{MPV}(d, k) = \frac{1}{\binom{d-1}{k-1}} \max_{\substack{0 \leq r \leq d-k \\ r \equiv d-k \pmod{2}}} \binom{d-1}{r}.$$

In particular,

$$\text{MPV}(d, d) = 1.$$

For every fixed $k \geq 2$, the growth is exponential in d :

$$\text{MPV}(d, k) \sim \frac{(k-1)!}{\sqrt{2\pi}} \frac{2^d}{d^{k-\frac{1}{2}}}, \quad d \rightarrow \infty.$$

Example: choosing the optimal index

Let $d = 10$ and $k = 3$. Then

$$0 \leq r \leq d - k = 7, \quad r \equiv d - k - 1 \equiv 0 \pmod{2}.$$

The admissible indices are

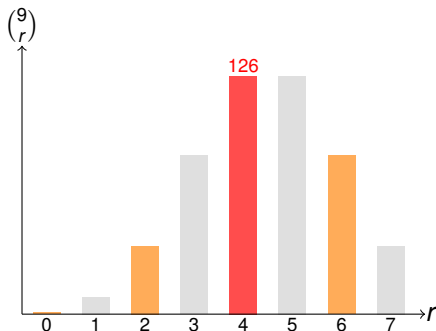
$$r \in \{0, 2, 4, 6\}.$$

The closest one to the peak of the binomial coefficients is

$$r_0 = 4.$$

Hence

$$\text{MNV}(10, 3) = -\frac{\binom{9}{4}}{\binom{9}{2}} = -\frac{7}{2}.$$



Example: the truncated regime

If $d - k \leq \lfloor \frac{d-1}{2} \rfloor$, the formula simplifies to

$$\text{MNV}(d, k) = -\frac{d - k}{k}.$$

For $d = 10$ and $k = 6$,

$$d - k = 4 \leq \left\lfloor \frac{d-1}{2} \right\rfloor, \quad r_0 = d - k - 1 = 3.$$

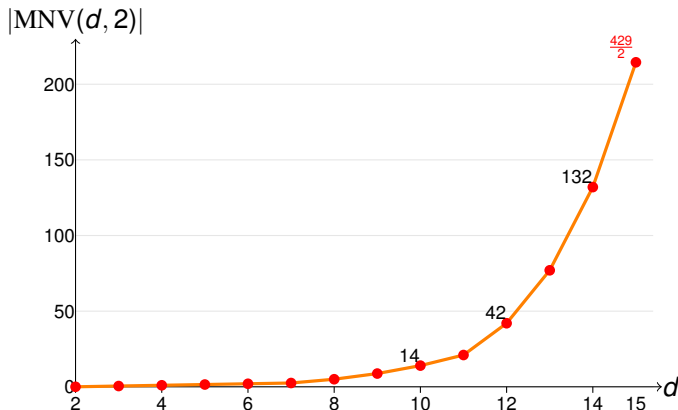
Therefore

$$\text{MNV}(10, 6) = -\frac{\binom{9}{3}}{\binom{9}{5}} = -\frac{84}{126} = -\frac{2}{3}.$$

Remark: Although the maximal positive volume equals 1, the negative value $\text{MNV}(10, 6) < 0$ shows that 6-increasing quasi-copulas are not necessarily copulas.

Growth of the maximal negative volume

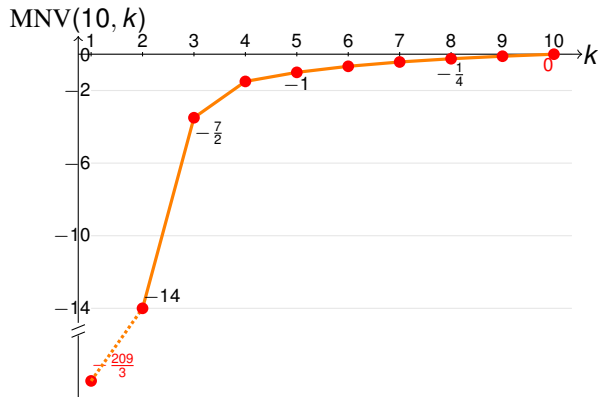
Values of $|\text{MNV}(d, 2)|$ for $2 \leq d \leq 15$:



For fixed $k = 2$, the optimal index approaches the peak of $\binom{d-1}{r}$, and the magnitude of the negative volume grows rapidly with the dimension.

Effect of increasing k for fixed dimension

For $d = 10$:



$$MNV(10, 1) = -\frac{209}{3},$$

$$MNV(10, 2) = -14,$$

$$MNV(10, 3) = -\frac{7}{2},$$

$$MNV(10, 5) = -1,$$

$$MNV(10, 8) = -\frac{1}{4},$$

$$MNV(10, 10) = 0.$$

As k increases, the maximal negative volume approaches zero.

$$MNV(d, d) = 0$$

Behavior for fixed d and increasing k

The classes of k -increasing quasi-copulas become smaller as k increases. Consequently,

$$\text{MNV}(d, k) \leq \text{MNV}(d, k+1) \leq 0,$$

so $|\text{MNV}(d, k)|$ decreases with k .

Small values of k

For

$$2 \leq k \leq \left\lfloor \frac{d}{2} \right\rfloor,$$

the maximizing binomial coefficient lies at or near the center:

$$|\text{MNV}(d, k)| \approx \frac{\binom{d-1}{\lfloor (d-1)/2 \rfloor}}{\binom{d-1}{k-1}}.$$

Hence

$$\frac{|\text{MNV}(d, k+1)|}{|\text{MNV}(d, k)|} \approx \frac{k}{d-k}.$$

The magnitude initially decreases rapidly.

Large values of k

For

$$k \geq \left\lfloor \frac{d}{2} \right\rfloor + 1,$$

the formula simplifies exactly to

$$\text{MNV}(d, k) = -\frac{d-k}{k}.$$

Writing $k = d - s$ gives

$$\text{MNV}(d, d-s) = -\frac{s}{d-s} \sim -\frac{s}{d} \quad (s \ll d).$$

Thus, near $k = d$, the magnitude decreases approximately linearly with the gap $d - k$.

Previous result: the case $k = 1$

Theorem (Omladič, Vuk, Z., 2026a)

Let $d \geq 7$ and $m = \lfloor d/2 \rfloor$.

Binomial coefficients

Set $c_0 = 0$ and, for $1 \leq i \leq m$,

$$c_i = \begin{cases} \binom{d-1}{i-1}, & d \text{ even,} \\ \binom{d-1}{2 \lfloor (i-1)/2 \rfloor + 1}, & d \text{ odd.} \end{cases}$$

Optimization

Define

$$w_{i,-} = \frac{\sum_{j=0}^{i-1} c_{m-j} - 1}{i+1}, \quad 1 \leq i \leq m,$$

and choose the first maximizer

$$i_0 = \min \left(\arg \max_{1 \leq i \leq m} w_{i,-} \right).$$

Then

$$\text{MNV}(d, 1) = - \max_{1 \leq i \leq m} w_{i,-} = -w_{i_0,-}.$$

An extremal box is

$$\mathcal{B} = \left[\frac{i_0}{i_0 + 1}, 1 \right]^d.$$

The dimensions $2 \leq d \leq 6$ are treated separately.

The cases $k = 1$ and $k \geq 2$

After symmetry reduction, let q_i be the common value at vertices with i upper coordinates, and let $h = b - a$. Define

$$\delta_i := \delta_i^{(1)} = q_i - q_{i-1}, \quad \delta_i^{(j)} = \delta_i^{(j-1)} - \delta_{i-1}^{(j-1)} \quad (j \geq 2).$$

	$k = 1$	$k \geq 2$
Constraints	The increments are independent: $0 \leq \delta_i \leq h.$	Higher differences satisfy $\delta_i^{(j)} \geq 0, \quad 1 \leq j \leq k.$ In particular, $0 \leq \delta_1 \leq \dots \leq \delta_d \leq h$.
Geometry	For $h = 1$, the feasible increments form the cube $[0, 1]^d$, with 2^d binary vertices.	For $k = 2$, they form a simplex with vertices $00 \dots 0011 \dots 11.$ Higher k gives a simplicial structure in the higher differences.
Dual	The d upper bounds remain separate, producing dual variables $l_1, \dots, l_d.$	Most constraints become redundant; the reduced dual has only four variables.
Extremizer	Several increments may be active. The answer involves partial binomial sums and a stopping index i_0.	Only one kth difference is active. The problem reduces to maximizing one binomial coefficient.

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Extremizer	Several increments may be active. The answer involves partial binomial sums and a stopping index i_0.	Only one kth difference is active. The problem reduces to maximizing one binomial coefficient.

For $k \geq 2$, the higher-difference constraints remove most of the combinatorial freedom.

Sketch of the proof

Proof idea I: reduction to vertex values

Fix

$$\mathcal{B} = \prod_{i=1}^d [a_i, b_i].$$

For $\mathbb{I} = (\mathbb{I}_1, \dots, \mathbb{I}_d) \in \{0, 1\}^d$, define

$$(x_{\mathbb{I}})_j = a_j + (b_j - a_j)\mathbb{I}_j, \quad q_{\mathbb{I}} := Q(x_{\mathbb{I}}).$$

The volume depends only on the 2^d vertex values:

$$V_Q(\mathcal{B}) = \sum_{\mathbb{I} \in \{0,1\}^d} (-1)^{d - \|\mathbb{I}\|_1} q_{\mathbb{I}}.$$

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$$V_Q(\mathcal{B}) = \sum_{\mathbb{I} \in \{0,1\}^d} (-1)^{d - \|\mathbb{I}\|_1} q_{\mathbb{I}}.$$

The problem becomes a finite linear program:

$$0 \leq q_{\mathbb{J}} - q_{\mathbb{I}} \leq b_{\ell} - a_{\ell} \quad \text{if } \mathbb{I} \prec_{\ell} \mathbb{J},$$

$$V_Q(F) \geq 0 \quad \text{for every } j\text{-face } F, 1 \leq j \leq d,$$

$$\max \left\{ 0, \sum_{j=1}^d (x_{\mathbb{I}})_j - d + 1 \right\} \leq q_{\mathbb{I}} \leq \min_j (x_{\mathbb{I}})_j \quad \text{for every } \mathbb{I}.$$

Here $\mathbb{I} \prec_{\ell} \mathbb{J}$ means that $\mathbb{I}$ and $\mathbb{J}$ differ only in coordinate ℓ , with $\mathbb{I}_{\ell} = 0$ and $\mathbb{J}_{\ell} = 1$.

Proof idea II: extension from the grid

For the optimizer constructed below, we will have

$$b_1 = \cdots = b_d = 1.$$

Prescribe $q_{\mathbb{I}}$ on

$$D = \prod_{i=1}^d \{a_i, 1\},$$

and set $Q = 0$ whenever some coordinate is zero.

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and set $Q = 0$ whenever some coordinate is zero.

The grid partitions $[0, 1]^d$ into boxes

$$\mathcal{B}_{\mathbb{I}} = \prod_{j=1}^d [\mathbb{I}_j a_j, a_j + \mathbb{I}_j(1 - a_j)], \quad \mathbb{I} \in \{0, 1\}^d.$$

Define a constant density on each box:

$$\rho|_{\text{int}(\mathcal{B}_{\mathbb{I}})} = \frac{V_Q(\mathcal{B}_{\mathbb{I}})}{\text{vol}(\mathcal{B}_{\mathbb{I}})}.$$

Then extend the vertex data by

$$Q(x) = \int_0^{x_1} \dots \int_0^{x_d} \rho(t) dt_d \dots dt_1.$$

The face constraints and multilinearity imply that this extension is a k -increasing quasi-copula.

Proof idea III: symmetry and finite differences

Averaging over all coordinate permutations gives an optimal symmetric solution:

$$a_i = a, \quad b_i = b, \quad q_{\mathbb{I}} = q_{\|\mathbb{I}\|_1}.$$

Hence only $d + 1$ vertex values remain, and

$$V_Q(\mathcal{B}) = \sum_{i=0}^d (-1)^{d-i} \binom{d}{i} q_i.$$

Proof idea III: symmetry and finite differences

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Hence only $d + 1$ vertex values remain, and

$$V_Q(\mathcal{B}) = \sum_{i=0}^d (-1)^{d-i} \binom{d}{i} q_i.$$

Define the successive backward differences

$$\delta_i^{(1)} = q_i - q_{i-1}, \quad \delta_i^{(j)} = \delta_i^{(j-1)} - \delta_{i-1}^{(j-1)}.$$

Then the increasing conditions reduce to

$$\delta_i^{(j)} \geq 0 \quad (1 \leq j \leq k, j \leq i \leq d).$$

After eliminating the redundant variables, the objective is

$$V_Q(\mathcal{B}) = \sum_{i=k}^d (-1)^{d+i} \binom{d-k}{i-k} \delta_i^{(k)}.$$

Proof idea IV: the symmetric primal program

Variables:

$$x_i := \delta_i^{(\min\{i,k\})}, \quad i = 1, \dots, d.$$

$$\text{MNV}(d, k) = \min \sum_{i=k}^d (-1)^{d+i} \binom{d-k}{i-k} x_i$$

subject to

$$b \leq 1,$$

$$a - b + \sum_{i=1}^d \alpha_i^{(k)} x_i \leq 0,$$

$$-a + q_0 + \sum_{i=1}^{d-1} \beta_i^{(k)} x_i \leq 0,$$

$$db - q_0 - \sum_{i=1}^d \gamma_i^{(k)} x_i \leq d - 1,$$

$$a, b, q_0, x_1, \dots, x_d \geq 0.$$

The coefficients $\alpha_i^{(k)}$, $\beta_i^{(k)}$, $\gamma_i^{(k)}$ have explicit formulas in terms of binomial coefficients.

The four inequalities give the four nonnegative dual variables y_1, y_2, y_3, y_4 .

Proof idea V: the explicit dual program

$$\begin{array}{ll}\text{maximize} & -y_1 - (d-1)y_4 \\ & y_1, y_2, y_3, y_4\end{array}$$

subject to

$$y_1 - y_2 + dy_4 \geq 0,$$

$$y_2 - y_3 \geq 0,$$

$$y_3 - y_4 \geq 0,$$

$$y_2 + (d-1)y_3 - dy_4 \geq 0,$$

$$\alpha_i^{(k)} y_2 + \beta_i^{(k)} y_3 - \gamma_i^{(k)} y_4 \geq 0, \quad 2 \leq i \leq k-1,$$

$$\alpha_i^{(k)} y_2 + \beta_i^{(k)} y_3 - \gamma_i^{(k)} y_4 \geq (-1)^{d+1-i} \binom{d-k}{i-k}, \quad k \leq i \leq d,$$

$$y_1, y_2, y_3, y_4 \geq 0.$$

Here $\alpha_i^{(k)}, \beta_i^{(k)}, \gamma_i^{(k)}$ are the coefficients from the primal program.

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$$y_2 - y_3 \geq 0,$$

$$y_3 - y_4 \geq 0,$$

$$y_2 + (d-1)y_3 - dy_4 \geq 0,$$

$$\alpha_i^{(k)} y_2 + \beta_i^{(k)} y_3 - \gamma_i^{(k)} y_4 \geq 0, \quad 2 \leq i \leq k-1,$$

$$\alpha_i^{(k)} y_2 + \beta_i^{(k)} y_3 - \gamma_i^{(k)} y_4 \geq (-1)^{d+1-i} \binom{d-k}{i-k}, \quad k \leq i \leq d,$$

$$y_1, y_2, y_3, y_4 \geq 0.$$

Here $\alpha_i^{(k)}, \beta_i^{(k)}, \gamma_i^{(k)}$ are the coefficients from the primal program.

By strong duality, the optimal value equals $\text{MNV}(d, k)$.

Proof idea VI: solving the dual

Choose

$$r_0 \in \arg \max_{\substack{0 \leq r \leq d-k \\ r \equiv d-k-1 \pmod{2}}} \binom{d-1}{r},$$

and set

$$W^* := \frac{\binom{d-1}{r_0}}{\binom{d-1}{k-1}}.$$

For $i = k + r$, we use

$$\gamma_{k+r}^{(k)} = \binom{d-r-1}{k-1}, \quad \frac{\binom{d-k}{r}}{\gamma_{k+r}^{(k)}} = \frac{\binom{d-1}{r}}{\binom{d-1}{k-1}}.$$

Hence W^* satisfies all the indexed dual inequalities.

A concrete optimal dual solution is

$$(y_1^*, y_2^*, y_3^*, y_4^*) = (W^*, W^*, W^*, 0).$$

Its objective value is

$$-y_1^* - (d-1)y_4^* = -W^*.$$

Therefore

$$\text{MNV}(d, k) = -W^*.$$

Proof idea VII: an optimal primal solution

Let

$$i_0 := k + r_0, \quad G := \gamma_{i_0}^{(k)} = \binom{d-1-r_0}{k-1}.$$

Complementary slackness gives an optimal primal solution with

$$\begin{aligned} b^* &= 1, & a^* &= \frac{k-1}{d-1-r_0}, & q_0^* &= 0, \\ x_{i_0}^* &= \frac{1}{G}, & x_i^* &= 0 \quad (i \neq i_0). \end{aligned}$$

Thus only one k th difference is active. Solving the corresponding finite-difference equation gives

$$q_i^* = \frac{\binom{i-r_0-1}{k-1}}{\binom{d-1-r_0}{k-1}}, \quad 0 \leq i \leq d,$$

where $\binom{n}{m} = 0$ for $n < m$.

For

$$\mathcal{B} = [a^*, 1]^d, \quad Q(z) = q_{d-S(z)}^*, \quad z \in \text{Vert}(\mathcal{B}),$$

these vertex values extend to a k -increasing quasi-copula and

$$V_Q(\mathcal{B}) = -W^* = \text{MNV}(d, k).$$

Future work

1. Investigate extreme volumes for **ultramodular and modular quasi-copulas**.
2. Characterize the **extreme points** of the convex set of k -increasing quasi-copulas.

The maximal-volume quasi-copula constructed in the paper is symmetric and therefore is not an extreme point.

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Thank you for your attention!